Brent Oil Price Change Point Analysis – Task 1 Interim Report

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Introduction

This report documents the completion of Task 1 of the Brent Oil Price Change Point Analysis project at Birhan Energies. The primary objective of Task 1 is to establish a strong data science foundation and prepare a curated dataset containing historical oil prices with meaningful structural patterns for subsequent modeling and change point detection.

2. Data Source and Preparation

For this task, we extracted historical Brent crude oil prices from publicly available records, covering the period starting from May 1987. The data was cleaned, formatted with appropriate datetime objects, and stored in a structured CSV file. An extract of the sample records is provided below.

Date	Price
20-May-87	18.63
21-May-87	18.45
22-May-87	18.55
25-May-87	18.6
26-May-87	18.63
27-May-87	18.6
28-May-87	18.6
29-May-87	18.58
01-Jun-87	18.65
02-Jun-87	18.68
03-Jun-87	18.75
04-Jun-87	18.78
05-Jun-87	18.65
08-Jun-87	18.75

3. Workflow Summary

The workflow for this task followed a structured data science approach inspired by several references:

- Data acquisition and preprocessing
- Exploratory data review
- Formatting for analysis readiness

Key references include:
https://www.datascience-pm.com/data-science-workflow/
https://towardsdatascience.com/mastering-the-data-science-workflow-2a47d8b613c4
https://www.knowledgehut.com/blog/data-science/data-science-workflow
https://medium.com/codex/a-comprehensive-guide-to-master-the-data-science-workflow-739295117d67

4. Output and Directory Structure

The final deliverable of this task is the `brent_oil_events.csv`, generated using a reproducible Python script. The project is version-controlled on GitHub and follows a clean directory structure:

```
brent-oil-change-point-analysis/
├── data/
│   └── brent_oil_events.csv
├── scripts/
│   └── generate_event_csv.py
├── reports/
│   └── Task1_Report_Brent_Oil_Analysis.docx
├── .github/
│   └── workflows/
│       └── ci.yml
├── README.md
```

Assumptions and Limitations

The analysis assumes that geopolitical and economic events have an observable impact on Brent oil prices, but we acknowledge the fundamental limitation that correlation in time does not equate to causation. Even if price shifts follow major events, this does not prove those events caused the changes. Market reactions are influenced by numerous variables, including investor sentiment, pre-existing trends, and concurrent events.

Another key assumption is that historical event timing is accurate and well-documented. However, reporting lags or media interpretation can introduce uncertainty. Further, external shocks (e.g., wars, policy changes) may have delayed or nonlinear effects.

Limitations also include the granularity of the dataset—daily prices may miss intraday fluctuations, and event impacts may span multiple days. Bayesian change point models help detect structural breaks but are sensitive to hyperparameters and priors.

Communication Channels and Formats

To ensure stakeholders understand the findings, results will be communicated via:
- Visual dashboards (e.g., time series plots with change points marked) using tools like Plotly and/or Streamlit.
- Executive summary reports (PDF) tailored for non-technical stakeholders.
- GitHub repository hosting code, structured data, and documentation.

Visual clarity and interpretability will be prioritized, especially when presenting probabilistic results from Bayesian models.

Understanding the Model and Data

Key references on data science workflow, Bayesian statistics, and change point detection were reviewed to guide this analysis. Models will be implemented in PyMC3, which uses Bayesian inference and Markov Chain Monte Carlo (MCMC) methods.

We conducted a preliminary investigation into Brent oil price time series characteristics. Visual inspection of the dataset indicates long-term trends with periods of volatility and stability. Stationarity tests (e.g., Augmented Dickey-Fuller) will be applied in later steps to confirm assumptions. These properties inform the modeling approach, especially regarding prior selection and likelihood choice in Bayesian frameworks.

Change point models are used to identify points in time where the statistical properties of the time series shift—e.g., mean, variance, or slope. This is crucial in understanding when structural changes occurred and associating them (tentatively) with global events. The outputs of such models include posterior probabilities of change points, estimated dates, and parameters (e.g., average price levels) before and after breaks.

Despite their strengths, change point models can misidentify breaks in the presence of noise or gradual trends, and do not inherently explain *why* the change occurred—that must be inferred by correlating with external data.

Conclusion

This first part of the project helped us understand the Brent oil price data and plan how we will analyze changes in it. We reviewed useful resources, organized the data, and prepared to use a Bayesian model to detect key shifts in oil prices that may be linked to political or economic events.

We also discussed the difference between finding patterns in data and proving actual causes, which is important when sharing results. Our goal in the next steps is to build the model, find important change points, and communicate the findings clearly to both technical and non-technical teams at Birhan Energies.